

Exp No: 15

Experiment to Understand Data Preprocessing in Data Science

Aim

To understand the importance of data preprocessing and perform data cleaning, missing value handling, categorical encoding and label encoding on the given dataset.

Program Description

The program loads the dataset and handles missing values in the **Country, Age and Salary** columns. Categorical values in Country are converted into dummy variables, and the Purchased column is converted into numerical values using label encoding.

Program

```
import numpy as np

import pandas as pd

df=pd.DataFrame({
'Country':['France','Spain','Germany','Spain','Germany','France','Spain',np.nan,'France','France'],
'Age':[44,27,30,38,40,35,np.nan,48,50,37],
'Salary':[72000,48000,54000,61000,np.nan,58000,52000,79000,83000,67000],
'Purchased':['No','Yes','No','No','Yes','Yes','No','Yes','No','Yes']
})

print(df)

print(df.info())

print(df.Country.mode())

print(df.Country.mode()[0])

print(type(df.Country.mode()))

df.Country=df.Country.fillna(df.Country.mode()[0])

df.Age=df.Age.fillna(df.Age.median())

df.Salary=df.Salary.fillna(round(df.Salary.mean()))

print(df)

print(pd.get_dummies(df.Country))
```

```

updated_dataset=pd.concat([pd.get_dummies(df.Country),df.iloc[:,[1,2,3]]],axis=1)

print(updated_dataset)

print(df.info())

updated_dataset.Purchased=updated_dataset.Purchased.replace(['No','Yes'],[0,1])

print(updated_dataset)

```

Output:

```

   Country  Age  Salary Purchased
0  France  44.0  72000.0         No
1   Spain  27.0  48000.0         Yes
2  Germany  30.0  54000.0         No
3   Spain  38.0  61000.0         No
4  Germany  40.0     NaN         Yes
5  France  35.0  58000.0         Yes
6   Spain  NaN  52000.0         No
7    NaN  48.0  79000.0         Yes
8  France  50.0  83000.0         No
9  France  37.0  67000.0         Yes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Country     9 non-null      object
1   Age         9 non-null      float64
2   Salary      9 non-null      float64
3   Purchased   10 non-null     object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes
None
0   France
Name: Country, dtype: object
France
<class 'pandas.core.series.Series'>
   Country  Age  Salary Purchased
0  France  44.0  72000.0         No
1   Spain  27.0  48000.0         Yes
2  Germany  30.0  54000.0         No
3   Spain  38.0  61000.0         No
4  Germany  40.0  63778.0         Yes
5  France  35.0  58000.0         Yes

```

```

6    Spain  38.0  52000.0    No
7    France 48.0  79000.0    Yes
... 8    France 50.0  83000.0    No
9    France 37.0  67000.0    Yes

```

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	No
1	False	False	True	27.0	48000.0	Yes
2	False	True	False	30.0	54000.0	No
3	False	False	True	38.0	61000.0	No
4	False	True	False	40.0	63778.0	Yes
5	True	False	False	35.0	58000.0	Yes
6	False	False	True	38.0	52000.0	No
7	True	False	False	48.0	79000.0	Yes
8	True	False	False	50.0	83000.0	No
9	True	False	False	37.0	67000.0	Yes

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Country     10 non-null     object
1   Age         10 non-null     float64
2   Salary      10 non-null     float64

```

```

1   Age         10 non-null     float64
2   Salary      10 non-null     float64
3   Purchased   10 non-null     object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes
None

```

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	True	False	False	50.0	83000.0	0
9	True	False	False	37.0	67000.0	1

```

/tmp/ipykernel_608/997867076.py:35: FutureWarning: Downcasting behavior in 'replace' is deprecated and will be removed in a future version. To retain the old behavior, explicitly call 'result.in
updated_dataset.Purchased=updated_dataset.Purchased.replace(['No','Yes'],[0,1])

```

Result:

Thus, the given dataset was successfully preprocessed by handling missing values and converting categorical data into numerical form.